



Edexcel International A Level Maths: Pure 1



Your notes

1.2 Quadratics

Contents

- * 1.2.1 Quadratic Graphs
- * 1.2.2 Discriminants
- * 1.2.3 Completing the square
- * 1.2.4 Solving Quadratic Equations
- * 1.2.5 Further Solving Quadratic Equations (Hidden Quadratics)

1.2.1 Quadratic Graphs

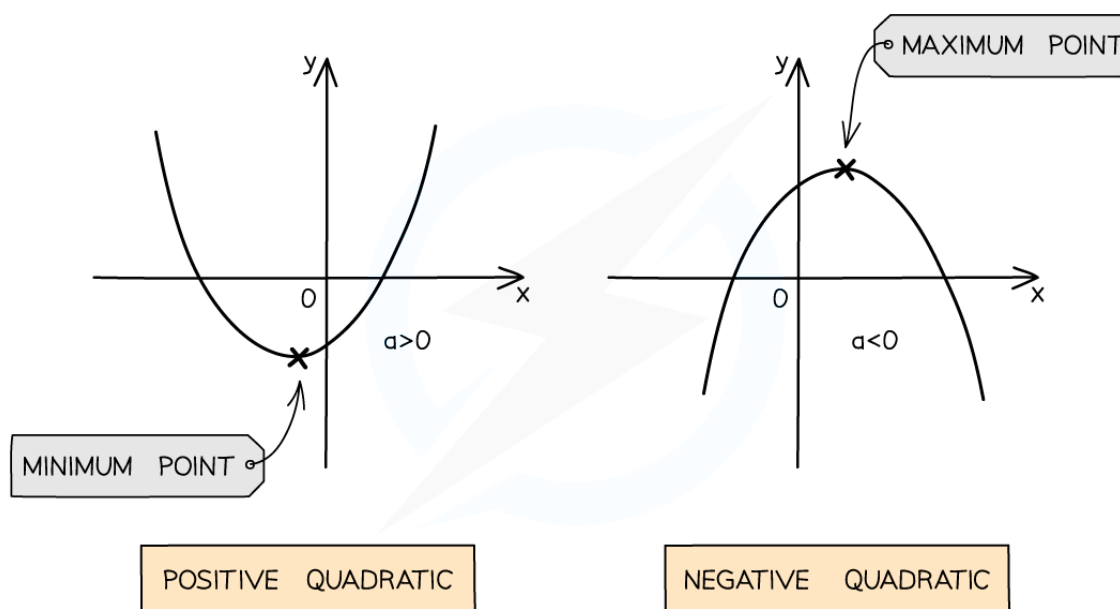


Your notes

Quadratic Graphs

What are quadratic graphs?

- The general equation of a quadratic graph is $y = ax^2 + bx + c$
- Their shape is called a parabola ("U" shape)
- Positive quadratics have a value of $a > 0$ so the parabola is upright U
- Negative quadratics have a value of $a < 0$ so the parabola is upside down $\cap$



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Using quadratic graphs

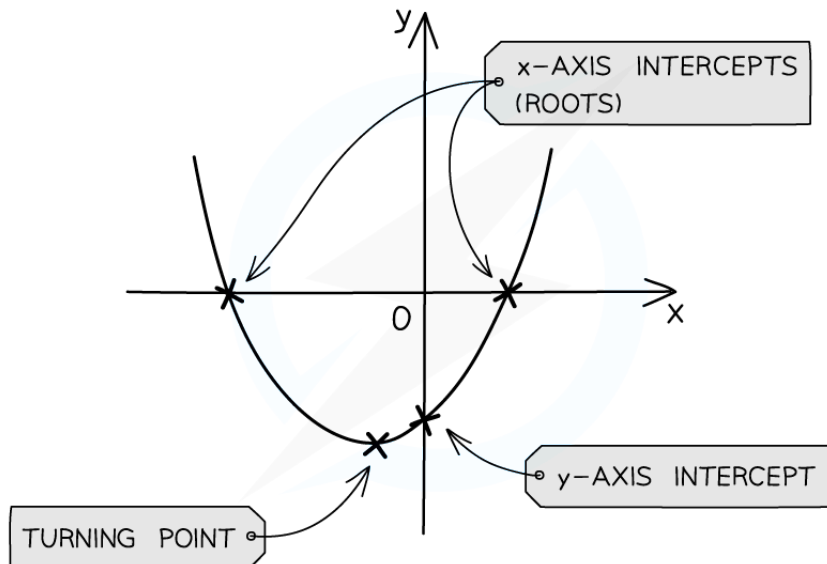
You need to be able to:

- **sketch** a quadratic graph given an equation or information about the graph
- determine, from the equation, the **axes intercepts**
- **factorise**, if possible, to find the roots of the quadratic function
- find the coordinates of the **turning point** (maximum or minimum)

You may have to rearrange the equation before you can find some of these things



Your notes



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Examiner Tip

- Your calculator may tell you the **roots** of a quadratic function and the coordinates of the turning point
- But don't rely on it – think about how many marks the question is worth and how much method/working you should show
- Remember sometimes you'll need to rearrange an equation into the form $y = ax^2 + bx + c$

 Worked example



Your notes



Your notes



Sketch the graph of $y = 2x^2 + 5x - 12$ stating the coordinates of the axes intercepts and the turning point.

$$y = 2x^2 + 5x - 12$$

POSITIVE QUADRATIC: U-SHAPED

$$\text{AT } x=0, y=-12$$

SET $x=0$ FOR y -AXIS INTERCEPT

$$y = (2x-3)(x+4)$$

FACTORISE AND SOLVE FOR x -AXIS INTERCEPTS

$$y=0 \text{ AT } x=-\frac{3}{2} \text{ AND } x=-4$$

∴ AXES INTERCEPTS ARE $(0, -12)$, $(-4, 0)$ AND $(-\frac{3}{2}, 0)$

$$y = 2(x^2 + \frac{5}{2}x) - 12$$

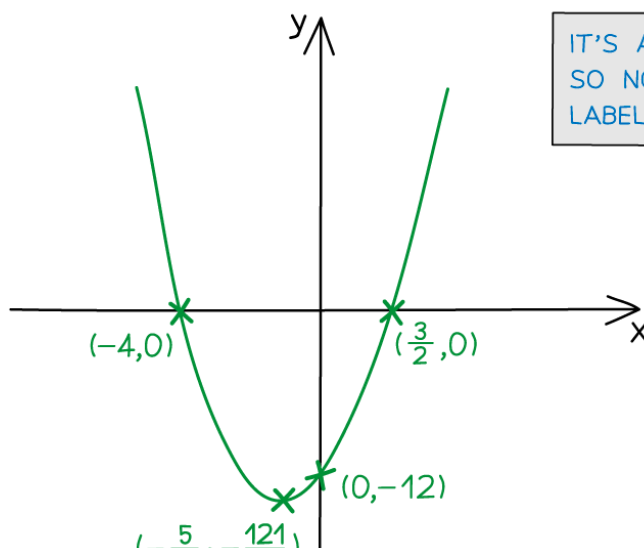
COMPLETE THE SQUARE...

$$y = 2[(x + \frac{5}{4})^2 - \frac{25}{16}] - 12$$

$$y = 2(x + \frac{5}{4})^2 - \frac{121}{8}$$

...TO FIND THE TURNING POINT

∴ TURNING POINT IS $(-\frac{5}{4}, -\frac{121}{8})$



IT'S A SKETCH
SO NO NEED TO
LABEL AXES

\ 4 ' 8 ' |



Your notes

1.2.2 Discriminants



Your notes

Discriminants

What is a discriminant?

- The discriminant is the part of the quadratic formula that is under the square root sign $b^2 - 4ac$
- It is sometimes denoted by the Greek letter Δ (capital delta)

How does the discriminant affect graphs and roots?

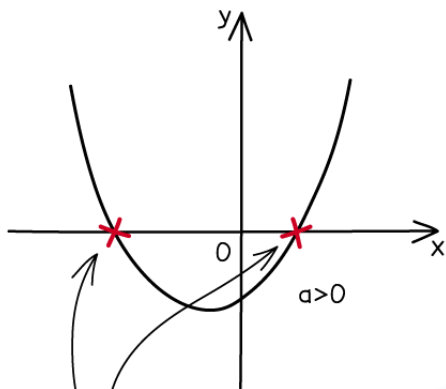
There are three options for the outcome of the discriminant:

- If $b^2 - 4ac > 0$ the quadratic crosses the x-axis **twice** meaning there are **two distinct real roots**
- If $b^2 - 4ac = 0$ the quadratic touches the x-axis **once** meaning there is **one real root** (also called repeated roots)
- If $b^2 - 4ac < 0$ the quadratic **does not cross** the x-axis meaning there are no real roots



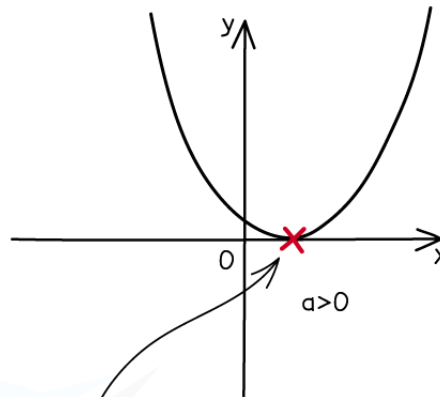
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IF $b^2 - 4ac > 0$



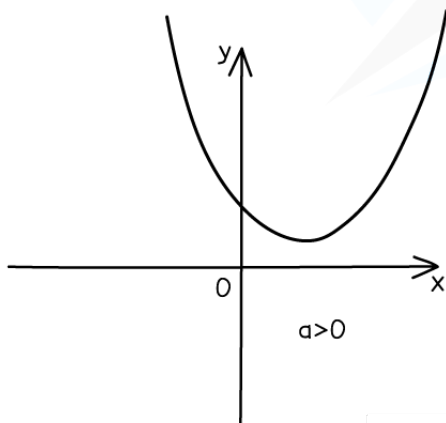
TWO DISTINCT
REAL ROOTS

IF $b^2 - 4ac = 0$



ONE REAL ROOT
(REPEATED ROOTS)

IF $b^2 - 4ac < 0$



NO REAL ROOTS

Discriminant and inequalities

- You need to be able to **set up** and **solve** equations and inequalities (often quadratic) arising from the discriminant
- Sketch the quadratic and decide whether you're looking above or below zero to write your solutions correctly



Your notes

IF
 $kx^2 + 3kx - 2k^2 = 0$

HAS REAL, DISTINCT ROOTS,

THEN " $b^2 - 4ac > 0$ "

$$(3k)^2 - 4(k)(-2k^2) > 0$$

$$9k^2 + 8k^3 > 0$$

ORIGINAL
QUADRATIC
IN x

INEQUALITY
ARISING FROM
THE DISCRIMINANT

Examiner Tip

- When questions just mention "real roots", the roots could be **distinct** or **repeated** (i.e. they aren't talking about complex numbers!)
- In these cases, you only need to worry about solving $b^2 - 4ac \geq 0$
- When solving using inequalities always sketch the quadratic and decide whether you're looking above or below zero to help write your solutions correctly

 Worked example



Your notes



Find the values of k for which the quadratic function

$f(x) = 2kx^2 + kx - k + 2$ has real roots.

$$f(x) = 2kx^2 + kx - k + 2$$

$$(k)^2 - 4(2k)(-k+2) \geq 0$$

" $b^2 - 4ac \geq 0$ "
FOR REAL ROOTS

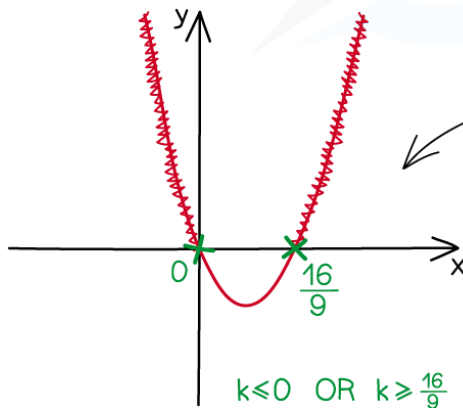
$$k^2 - 8k(2-k) \geq 0$$

$$k^2 - 16k + 8k^2 \geq 0$$

$$9k^2 - 16k \geq 0$$

$$k(9k - 16) \geq 0$$

YOUR CALCULATOR
MAY SOLVE THIS...



...BUT A SKETCH IS
ESSENTIAL

1.2.3 Completing the square



Your notes

Completing the square

What is completing the square?

- Completing the square is another method used to solve quadratic equations
- It simply means writing $y = ax^2 + bx + c$ in the form $y = a(x + p)^2 + q$
- It can be used to help find other information about the quadratic like coordinates of the turning point

How do I complete the square?

The method used will depend on the value of the coefficient of the x^2 term in $y = ax^2 + bx + c$

When $a = 1$

- p is half of the coefficient of b
- q is $c - p^2$

e.g. $y = x^2 + 8x - 2$

STEP 1: " $p = \frac{b}{2}$ " $p = \frac{8}{2} = 4$

STEP 2: " $q = c - p^2$ " $q = -2 - 4^2 = -18$

STEP 3: WRITE THE ANSWER!

$$y = (x + 4)^2 - 18$$

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When $a \neq 1$

- You first need to take a out as a factor of the x^2 and x terms
- Then continue as above



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e.g. $y = 4x^2 + 16x + 5$

NO NEED TO
INVOLVE "c"

STEP 1: FACTOR "a" ON RHS $y = 4[x^2 + 4x] + 5$

STEP 2: WORKING WITH $[x^2 + 4x]$ ONLY, " $p = \frac{b}{2}$ "
 $p = \frac{4}{2} = 2$

STEP 3: STILL WITH $[x^2 + 4x]$ ONLY, " $q = c - p^2$ "
 $q = 5 - 2^2 = -4$

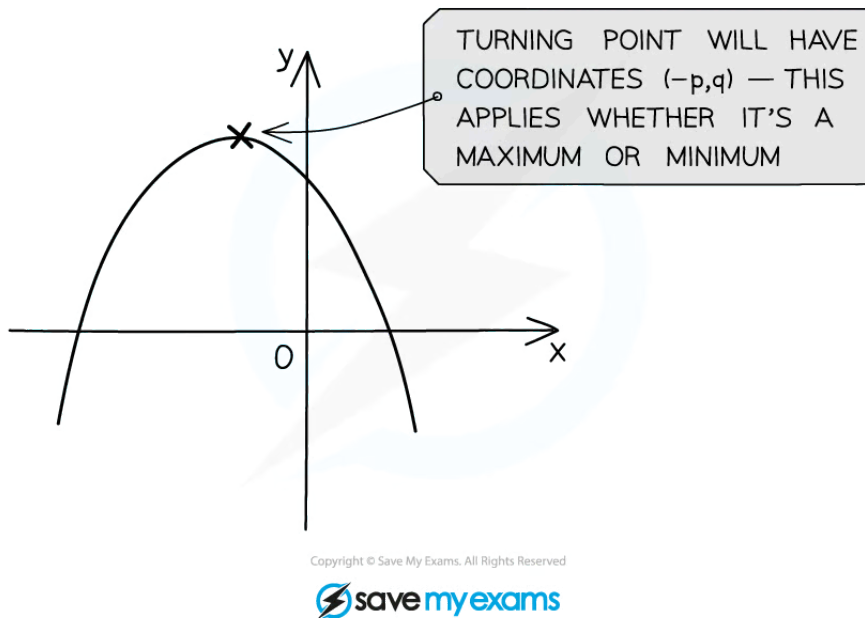
STEP 4: EXPAND AND SIMPLIFY $y = 4[(x+2)^2 - 4] + 5$
 $y = 4(x+2)^2 - 16 + 5$
 $y = 4(x+2)^2 - 11$

When is completing the square useful?

- Completing the square helps us find the **turning point** on a quadratic graph
- It can also help you **create the equation** of a quadratic when given the turning point



Your notes



- It can also be used to **prove** and/or **show** results using the fact that a squared term will always be greater than or equal to 0

e.g. $y = x^2 + 6x - 3$

$$y = (x+3)^2 - 12$$

$$\therefore y \geq -12$$

A SQUARED TERM
WILL HAVE A MINIMUM
VALUE OF 0

Examiner Tip

- Sometimes the question will explicitly ask you to complete the square
- Sometimes it will even remind you of the form to write it in
- But sometimes it will expect you to spot that completing the square is what you need to do to help with other parts of the question... like finding turning points!

 Worked example



Your notes



Your notes



(a) Show $2x^2 + 6x + k > 0$ for $k > \frac{9}{2}$

(b) What can you say about the roots of the equation

$$2x^2 + 6x + k = 0 \text{ when } k = \frac{9}{2}?$$

(a)

">0" IMPLIES A MINIMUM VALUE — SO
COMPLETE THE SQUARE WITH "a≠1"

$$2x^2 + 6x + k = 2[x^2 + 3x] + k$$

STEP 1: FACTOR "a"

$$= 2\left[\left(x + \frac{3}{2}\right)^2 - \frac{9}{4}\right] + k$$

STEP 2: "p = $\frac{b}{2}$ "

STEP 3: "q = c - p²"

NOTE WITH STEP 2 & STEP 3
YOU ARE ONLY WORKING WITHIN
THE SQUARE [] BRACKETS

$$= 2\left(x + \frac{3}{2}\right)^2 - \frac{9}{2} + k$$

$$= 2\left(x + \frac{3}{2}\right)^2 + k - \frac{9}{2}$$

STEP 4: EXPAND
AND SIMPLIFY

$$2\left(x + \frac{3}{2}\right)^2 \geq 0 \text{ FOR ALL VALUES OF } x$$

$$\therefore 2\left(x + \frac{3}{2}\right)^2 + k - \frac{9}{2} > 0 \text{ WHEN } k - \frac{9}{2} > 0$$

$$k > \frac{9}{2}$$

SHOW ALL STAGES
IN A "SHOW THAT"
QUESTION



Your notes

- (b) WHEN $k = \frac{9}{2}$ THE QUADRATIC BECOMES $2(x + \frac{3}{2})^2$ AND SO THERE WILL ONLY BE ONE ROOT (OR REPEATED ROOTS) TO THE EQUATION $2x^2 + 6x + k = 0$

THE EQUATION IS
NOW $2(x + \frac{3}{2})^2 = 0$

1.2.4 Solving Quadratic Equations



Your notes

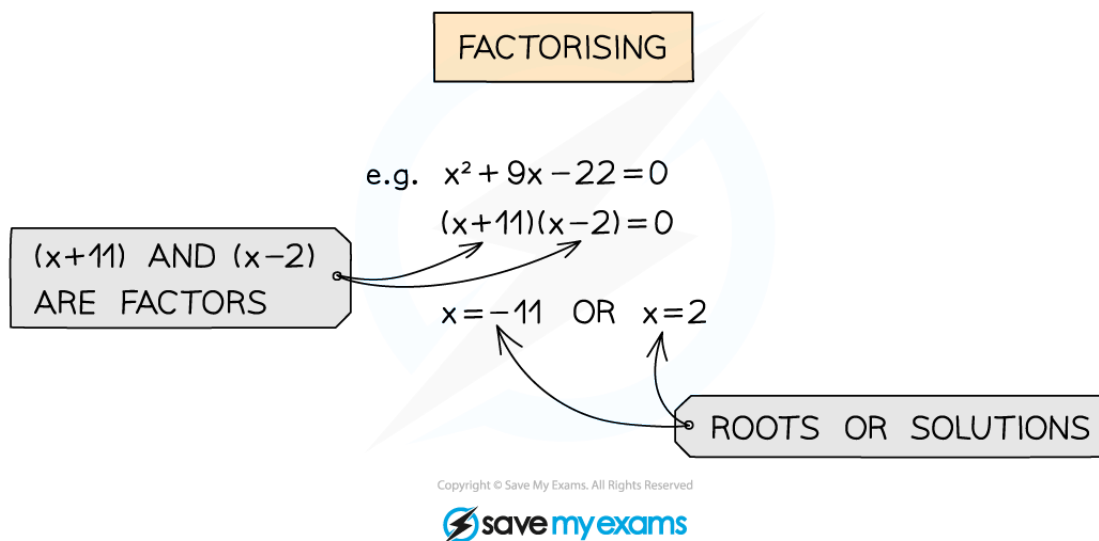
Solving Quadratic Equations

Solving quadratic equations

- We can solve quadratic equations when they are written in the form $ax^2 + bx + c = 0$
- If given an unusual looking equation, try to rearrange it into this form first
- The three ways to solve a quadratic you must know are
 - Factorising
 - Completing the square
 - Quadratic formula

Solving a quadratic equation by factorising

- Factorising is a great way to solve a quadratic quickly but won't work for all quadratics
- If the numbers are simple, try factorising first
- Once factorised, set each bracket to $= 0$ and solve



Solving a quadratic equation by completing the square

- Completing the square will work for any quadratic
- Make sure you know how to complete the square
- Remember this will help with questions involving turning points too



Your notes

COMPLETING THE SQUARE

e.g. $x^2 + 10x - 24 = 0$

COMPLETE
THE SQUARE:
"p = $\frac{b}{2}$ ", "q = c - p²"

$$(x+5)^2 - 49 = 0$$

$$(x+5)^2 = 49$$

$$x+5 = \pm 7$$

REMEMBER BOTH
POSITIVE AND
NEGATIVE
SQUARE ROOTS

$$x = 7 - 5 \quad \text{OR} \quad x = -7 - 5$$

$$x = 2 \quad \text{OR} \quad x = -12$$

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Solving a quadratic equation by the quadratic formula

- The quadratic formula might look complicated but it just uses the coefficients a, b and c from the quadratic equation
- The quadratic formula will work for any quadratic



Your notes

QUADRATIC FORMULA

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

e.g. $x^2 - 3x - 6 = 0$

DISCRIMINANT
FIRST...

" $b^2 - 4ac$ ": $(-3)^2 - 4(1)(-6) = 33$

... MAKES THIS
LINE EASIER!

$$x = \frac{3 \pm \sqrt{33}}{2}$$

$$x = \frac{3 + \sqrt{33}}{2} \quad \text{OR} \quad x = \frac{3 - \sqrt{33}}{2}$$

KEEP ANSWERS EXACT
UNLESS INSTRUCTED
OTHERWISE

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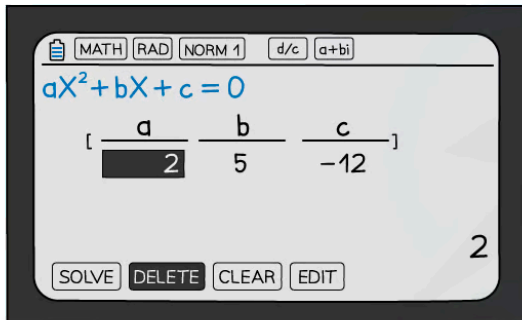


Solving a quadratic equation by using a calculator

- Most calculators now have the ability to solve quadratics
- Get used to how your calculator functions work



Your notes



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- So the solution to the quadratic equation $2x^2 + 5x - 12 = 0$ are $x = 1.5$ and $x = -4$

Examiner Tip

- A calculator can be super-efficient but be aware some marks are for method
- There will never be many marks for solving a quadratic at AS/A level
- Use your judgement:
 - is it a “show that” or “prove” question?
 - how many marks?
 - how long is the question?
- Remember the quadratic formula with a song... there are loads of fun ones on YouTube

 Worked example



Your notes

? Solve the equation $x + 5 = \frac{12}{x}$.

$$x + 5 = \frac{12}{x}$$

$$x^2 + 5x = 12$$

$$x^2 + 5x - 12 = 0$$

$$\begin{aligned} "b^2 - 4ac" &= 5^2 - 4 \times 1 \times (-12) \\ &= 25 + 48 \\ &= 73 \end{aligned}$$

$$\therefore x = \frac{-5 \pm \sqrt{73}}{2}$$

$$x = \frac{-5 + \sqrt{73}}{2}$$

$$x = \frac{-5 - \sqrt{73}}{2}$$

IT'S UNUSUAL SO TRY
REARRANGING.
*x TO GET RID OF
FRACTIONS

THIS DOESN'T
FACTORISE,
COMPLETING
THE SQUARE
IS NOT "NICE"
DUE TO THE 5

WORKING OUT
THE DISCRIMINANT
FIRST MAKES
THINGS EASIER
AND TIDY

CHECK THESE ON
YOUR CALCULATOR

1.2.5 Further Solving Quadratic Equations (Hidden Quadratics)



Your notes

Further Solving Quadratic Equations (Hidden Quadratics)

What are hidden quadratic equations?

- Hidden quadratic equations are quadratics written in terms of a function $f(x)$
- A normal quadratic appears in the form $ax^2 + bx + c = 0$
- Whereas a hidden quadratic appears in the form $a[f(x)]^2 + b[f(x)] + c = 0$
- This might look complicated but it simply means x has been replaced by $f(x)$
- e.g. $\sin^2 x + 2\sin x - 3 = 0$ is just the hidden quadratic of $x^2 + 2x - 3 = 0$ where $f(x) = \sin x$

How to solve hidden quadratic equations

- First rearrange the function into the form $a[f(x)]^2 + b[f(x)] + c = 0$
- Replacing the function and solving the 'normal' quadratic first
- Then substitute the function back into the solutions to solve the original quadratic



Your notes

e.g. $2\cos x = 3 - \cos^2 x$ $0^\circ \leq x \leq 360^\circ$

STEP 1: REARRANGE INTO THE FORM

$$a[f(x)]^2 + bf(x) + c = 0$$

$$\cos^2 x + 2\cos x - 3 = 0$$

STEP 2: • "IGNORE" THE FUNCTION OF x
AND SOLVE AS A NORMAL QUADRATIC
• USE A SUBSTITUTION, $y = f(x)$ IF
YOU PREFER

$$\begin{aligned} y = \cos x, \quad y^2 + 2y - 3 &= 0 \\ (y+3)(y-1) &= 0 \\ y = -3 \quad y = 1 \end{aligned}$$

STEP 3: USE THESE SOLUTIONS, AND THE
FUNCTION OF x , TO SOLVE THE
ORIGINAL QUESTION

$$\begin{array}{ll} \cancel{\cos x = -3} & \cos x = 1 \\ \text{NO SOLUTIONS} & x = 0^\circ, 360^\circ \end{array}$$

 Worked example



Your notes

? Solve the equation $3^{2x+1} = 82 \times 3^x - 27$.

$$3^{2x+1} - 82 \times 3^x + 27 = 0$$

REARRANGE SO IT "= 0" AND WORK TOWARDS CORRECT FORM

$$3^1 \times 3^{2x} - 82 \times 3^x + 27 = 0$$

AWKWARD TO SPOT BUT 82×3^x IS THE CLUE

$$3(3^x)^2 - 82(3^x) + 27 = 0$$

THIS IS NOW IN THE FORM $a[f(x)]^2 + bf(x) + c = 0$

LET $y = 3^x$

$$3y^2 - 82y + 27 = 0$$

$$(3y-1)(y-27) = 0$$

$$y = \frac{1}{3} \quad y = 27$$

YOU DO NOT HAVE TO USE A SUBSTITUTION

$$\therefore 3^x = \frac{1}{3} \quad 3^x = 27$$

$$x = -1 \quad x = 3$$

MAKE SURE YOU FIND x's FOR YOUR FINAL ANSWER